MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

ANTIMICROBIAL DOSING RECOMMENDATIONS FOR ADULT PATIENTS[1]

Antibiotics			
Amikacin IV	Ceftazidime IV	Daptomycin IV	Moxifloxacin IV/PO
Amoxicillin PO	Ceftazidime-avibactam IV	Doxycycline IV/PO	Omadacycline PO
Amoxicillin-clavulanate PO	Ceftolozane-tazobactam IV	Eravacycline IV	Omadacycline IV
Ampicillin IV	Ceftriaxone IV	Ertapenem IV	Oxacillin/nafcillin IV
Ampicillin-sulbactam IV	Cefuroxime IV	Fosfomycin PO	Nitrofurantoin capsules PO
Azithromycin IV/PO	Cefuroxime PO	Gentamicin IV	Nitrofurantoin suspension PO
Aztreonam IV	Cephalexin PO	Imipenem IV	Penicillin G IV
Aztreonam-avibactam IV	Ciprofloxacin IV	Imipenem-relebactam IV	Piperacillin-tazobactam IV
Cefazolin IV	Ciprofloxacin PO	Levofloxacin IV/PO	Polymyxin B IV
Cefepime IV	Clarithromycin PO	Linezolid IV/PO	Rifampin IV/PO
Cefiderocol IV	Clindamycin IV	Meropenem IV	Tigecycline IV
Cefoxitin IV	Clindamycin PO	Meropenem-vaborbactam IV	Tobramycin IV
Cefpodoxime PO	Colistin IV	Metronidazole IV/PO	Trimethoprim-sulfamethoxazole (TMP-SMX) IV/PO
Ceftaroline IV	Dalbavancin IV	Minocycline IV/PO	Vancomycin IV
Antifungals			
Amphotericin B Deoxycholate IV	Isavuconazole IV/PO	Posaconazole tablet IV/PO	Voriconazole PO
Liposomal Amphotericin B IV	Itraconazole PO	Posaconazole suspension PO	
Fluconazole IV/PO	Itraconazole PO (Tolsura)	Rezafungin IV	
Flucytosine PO	Micafungin IV	Voriconazole IV	
Antivirals			
Acyclovir IV	Ganciclovir IV	Oseltamivir PO	Valacyclovir PO
Acyclovir PO	Letermovir IV/PO	Peramivir IV	Valganciclovir PO
Baloxavir PO	Maribavir IV	Remdesivir IV	Zanamivir INH
Cidofovir IV	Molnupiravir PO	Ribavirin PO	
Foscarnet IV	Nirmatrelvir-ritonavir PO	Ribavirin PO Lung Transplant	
Footnotes			

ANTIBIOTICS	CrCl[1] > 50 mL/min	CrCl[1] 30-49 mL/min	CrCl[1] 10-29 mL/min	CrCl[1] < 10 mL/min; HD[2]	CRRT (rate 2.5 L/hr)[3]
AMIKACIN (IV)	[See Aminoglycoside Dosing in Adult Patients]				
AMOXICILLIN (PO)					
Uncomplicated cystitis (lower urinary tract infection)	500 mg q8h	500 mg q8h	500 mg q12h	500 mg q24h	500 mg q8h
Helicobacter pylori infection	1 g q12h	1 g q12h	500 mg q12h	500 mg q24h	1 g q12h
Pharyngitis	500 mg q12h	500 mg q12h	500 mg q12h	500mg q24h	500 mg q12h
Standard dosing for most infections	1 g q8h	1 g q8h	1 g q12h	1 g q24h	1 g q8h
Serious systemic infections, including bacteremia, pyelonephritis, osteomyelitis,	1 g q6h	1 g q8h	1 g q12h	1 g q24h	1 g q8h
AMOXICILLIN-CLAVULANATE (PO)					
Standard dosing for most infections, including community-acquired pneumonia, intra-abdominal infections	875 mg q12h	875 mg q12h	500 mg q12h	500 mg q24h	875 mg q12h
Serious systemic infections, including bacteremia, pyelonephritis, osteomyelitis	875mg q8h	875mg q8h	875mg q12h	875mg q24h	875mg q8h
AMPICILLIN (IV)					
Uncomplicated cystitis (lower urinary tract infection)	1 g q6h	1 g q8h	1 g q12h	1 g q24h	1 g q8h
Endocarditis; CNS infection	2 g q4h	2 g q6h	2 g q8h	2 g q12h	2 g q6h
All other infections	2 g q6h	2 g q8h	2 g q12h	2 g q24h	2 g q8h
AMPICILLIN-SULBACTAM (IV)					
Uncomplicated cystitis (lower urinary tract infection)	1.5 g q6h	1.5 g q8h	1.5 g q12h	1.5 g q24h	1.5 g q8h
Acinetobacter infection	3 g q4h	3 g q6h	3 g q8h	3 g q12h	3 g q6h
All other infections	3 g q6h	3 g q8h	3 g q12h	3 g q24h	3 g q8h
AZITHROMYCIN (IV/PO)					
Legionella or pneumonia in ICU	500 mg q24h				
Non-tuberculous mycobacterial infection	Consult infectious diseases				
All other infections	500 mg x1, then 250 mg q24h				
AZTREONAM (IV)					
CNS infections	2g q6h	2g q6h	2g q8h	2g q12h	2g q6h
All other infections	2 g q8h	2 g q8h	2 g q12h	1 g q12h	2 g q8h
AZTREONAM-AVIBACTAM (IV)	2.67 g (aztreonam 2 g - avibactam 0.67 g) x1, then 2 g (aztreonam 1.5 g - avibactam 0.5 g) q6h	2.67 g (aztreonam 2 g - avibactam 0.67 g) x1, then 1 g (aztreonam 0.75 g - avibactam 0.25 g) q6h	1.8 g (aztreonam 1.35 g - avibactam 0.45 g) x1, then 0.9 g (aztreonam 0.675 g - avibactam 0.225 g) q8h	1.33 g (aztreonam 1 g - avibactam 0.33 g) x1, then 0.9 g (aztreonam 0.675 g - avibactam 0.225 g) q12h	2.67 g (aztreonam 2 g - avibactam 0.67 g) x1, then 1 g (aztreonam 0.75 g - avibactam 0.25 g) q6h

ANTIBIOTICS	CrCl[1] > 50 mL/min	CrCl[1] 30-49 mL/min	CrCl[1] 10-29 mL/min	CrCl[1] < 10 mL/min; HD[2]	CRRT (rate 2.5 L/hr)[3]	
CEFAZOLIN (IV)						
Uncomplicated cystitis (lower urinary tract infection)	1 g q8h	1 g q8h	1 g q12h	500 mg q24h	1 g q8h	
All other infections	2 g q8h	2 g q8h	2 g q12h	1 g q24h or 2 g post HD 3x/week	2 g q8h	
CEFEPIME (IV)	**CrCl ≥ 90 mL/min**	**CrCl 50-89 mL/min**	**CrCl 30-49 mL/min**	**CrCl 10-29 mL/min**	**CrCl < 10 mL/min or iHD**	**CRRT (rate 2.5 L/hr)**
All infections except those noted below[6]	2 g q8h	1g q8h	1 g q12h	1 g q24h	2 g post HD 3x/week or 1 g q24h	2 g q12h
CNS infection, Fever and neutropenia in hematologic malignancy (CAR-T excluded)	2 g q8h	2 g q8h	2 g q12h	1 g q12h	2 g post HD 3x/week or 1 g q24h	2 g q12h

CEFIDEROCOL (IV)	CrCl > 120 mL/min	CrCl 60-119 mL/min	CrCl 30-59 mL/min	CrCl 15-29 mL/min	CrCl < 15 mL/min or iHD	CRRT
All indications	2 g q6h	2 g q8h	1.5 g q8h	1 g q8h	750 mg q12h	Refer to footnote 5

CEFOXITIN (IV)					
Systemic infection	2 g q6h	2 g q8h	2 g q12h	1 g q24h	2 g q8h
Atypical mycobacterial infection	3 g q6h	3 g q8h	3 g q12h	2 g q24h	3 g q8h
CEFPODOXIME (PO)					
Uncomplicated cystitis (lower urinary tract infection)	100 mg q12h	100 mg q12h	100 mg q24h	100 mg post HD 3x/week	100 mg q12h
Other infections, including bacteremia, pyelonephritis, pneumonia, and other serious systemic infections	400 mg q12h	400 mg q12h	400 mg q24h	400 mg post HD 3x/week	400 mg q12h
CEFTAROLINE (IV)					
Skin & soft tissue infection or cystitis (lower urinary tract infection)	600 mg q12h	400 mg q12h	300 mg q12h	200 mg q12h	400 mg q12h
Other MRSA infections, including bacteremia, endocarditis, and osteomyelitis	600 mg q8h	400 mg q8h	300 mg q8h	200 mg q8h	400 mg q8h
CEFTAZIDIME (IV)	2 g q8h	2 g q12h	1 g q12h	1 g q24h or 2 g post-HD 3x/week	2 g q12h
CEFTAZIDIME-AVIBACTAM (IV)	2.5 g (ceftazidime 2 g - avibactam 0.5 g) q8h	1.25 g (ceftazidime 1 g - avibactam 0.25 g) q8h	1.25g (ceftazidime 1 g - avibactam 0.25 g) q12h	1.25 g (ceftazidime 1 g - avibactam 0.25 g) q24h or 2.5 g (ceftazidime 2 g – avibactam 0.5 g) post HD 3x/week	1.25 g (ceftazidime 1 g - avibactam 0.25 g) q8h

ANTIBIOTICS	CrCl[1] > 50 mL/min	CrCl[1] 30-49 mL/min	CrCl[1] 10-29 mL/min	CrCl[1] < 10 mL/min; HD[2]	CRRT (rate 2.5 L/hr)[3]
CEFTAZIDIME-AVIBACTAM (IV) + AZTREONAM (IV)	2.5 g (ceftazidime 2 g - avibactam 0.5 g) q8h + aztreonam 2g q8h	1.25 g (ceftazidime 1 g - avibactam 0.25 g) q8h + aztreonam 2g q8h	1.25 g (ceftazidime 1g g - avibactam 0.25 g) q12h + aztreonam 2g q12h	1.25 g (ceftazidime 1g g - avibactam 0.25 g) q24h + aztreonam 2g q24h	1.25 g (ceftazidime 1 g - avibactam 0.25 g) q8h + aztreonam 2g q8h
CEFTOLOZANE-TAZOBACTAM (IV)					
Uncomplicated cystitis (lower urinary tract infection)	1.5 g q8h	750 mg q8h	375 mg q8h	750 mg x1, then 150 mg q8h	750 mg q8h
All other infections	3 g q8h	1.5 g q8h	750 mg q8h	2.25 g x1, then 450 mg q8h	1.5 g q8h
CEFTRIAXONE (IV)					
Uncomplicated cystitis (lower urinary tract infection)	1 g q24h				
CNS infections; enterococcal endocarditis (in combination with ampicillin)	2 g q12h				
All other infections, including community-acquired pneumonia	2 g q24h				
CEFUROXIME (IV)					
Uncomplicated cystitis (lower urinary tract infection)	750 mg q8h	750 mg q8h	750 mg q12h	750 mg q24h	750 mg q8h
All other infections	1.5 g q8h	1.5 g q8h	1.5 g q12h	1.5 g q24h	1.5 g q8h
CEFUROXIME (PO)					
Uncomplicated cystitis (lower urinary tract infection)	250 mg q12h	250 mg q12h	250 mg q24h	250 mg q48h	250 mg q12h
All other infections	500 mg q12h	500 mg q12h	500 mg q24h	500 mg q48h	500 mg q12h
CEPHALEXIN (PO)					
Uncomplicated cystitis (lower urinary tract infection), asymptomatic bacteriuria	1 g q12h	1 g q12h	500 mg q12h	500 mg q24h	1000 mg q12h
Skin-soft tissue infection, osteomyelitis due to streptococci	1 g q8h	1 g q8h	1 g q12h	500 mg q12h	1 g q8h
All other infections, including bacteremia, endocarditis, osteomyelitis due to MSSA, pyelonephritis, and other systemic infections	1 g q6h	500 mg q6h or 1 g q8h	500 mg q8h or 1 g q12h	500 mg q24h or 1 g q24h	500 mg q6h or 1 g q8h
CIPROFLOXACIN (IV)					
Uncomplicated cystitis (lower urinary tract infection)	400 mg q12h	400 mg q12h	400 mg q24h	200 mg q24h	400 mg q24h
All other infections	400 mg q8h	400 mg q8h	400 mg q12h	400 mg q24h	400 mg q8h

ANTIBIOTICS	CrCl[1] > 50 mL/min	CrCl[1] 30-49 mL/min	CrCl[1] 10-29 mL/min	CrCl[1] < 10 mL/min; HD[2]	CRRT (rate 2.5 L/hr)[3]	
CIPROFLOXACIN (PO)						
Uncomplicated cystitis (lower urinary tract infection)	500 mg q12h	500 mg q12h	500 mg q24h	250 mg q24h	250 mg q12h	
All other infections, including bacteremia, pyelonephritis, pneumonia, and other systemic infections	750 mg q12h	750 mg q12h	750 mg q24h	500 mg q24h	750 mg q12h	
CLARITHROMYCIN	500 mg 12h	500 mg q12h	500 mg q24h	500 mg q24h	500 mg q12h	
CLINDAMYCIN (IV)	600-900 mg q8h					
CLINDAMYCIN (PO)	300-450 mg q6-8h					
COLISTIN BASE (IV) *						
Severe infection	5 mg/kg x1 (max: 300 mg) then 2.5 mg/kg q12h	5 mg/kg x1 (max: 300 mg) then 1.75 mg/kg q12h	5 mg/kg x1 (max: 300 mg) then 2.5 mg/kg q24h	5 mg/kg x1 (max: 300 mg) then 1.5 mg/kg q24h	5 mg/kg x1 (max: 300 mg) then 2.5 mg/kg q12h	
DALBAVANCIN (IV)						
Skin & soft tissue infection	1.5 g x1 dose	1.5 g x1 dose	1.125 g x1 dose	1.5 g for patients on HD	No Data Available	
Bone & Joint infection	1.5 g given on day 1 and day 8	1.5 g given on day 1 and day 8	1.5 g given on day 1 and day 8	1.5 g given on day 1 and day 8	No Data Available	
DAPTOMYCIN (IV) *						
Enterococcus infections (excluding uncomplicated cystitis); *Staphylococcus aureus* infections	10 mg/kg q24h	10 mg/kg q24h	10 mg/kg q48h	10 mg/kg post HD 3x/week or q48h	10 mg/kg q24h	
All other infections	6 mg/kg q24h	6 mg/kg q24h	6 mg/kg q48h	6 mg/kg post HD 3x/week or q48h	6 mg/kg q24h	
DOXYCYCLINE (IV/PO)	200 mg x1, then 100 mg q12h					
ERAVACYCLINE (IV)	1 mg/kg q12h					
ERTAPENEM (IV)	1 g q24h	1 g q24h	500 mg q24h	500 mg q24h or 1000 mg post HD 3x/week (preferred for outpatients)	1 g q24h	
FOSFOMYCIN (PO)						
Cystitis (lower urinary tract infection)	3 g x1 dose					
Prostatitis (*E. coli* or *E. faecalis* only; in consultation with ID)	3 g daily x7 doses followed by 3 g q48h x6 – 12 weeks					
GENTAMICIN (IV)	[See Aminoglycoside Dosing in Adult Patients]					
IMIPENEM (IV)						
Nocardia and atypical mycobacteria	500 mg q6h	500 mg q8h	500 mg q12h	250 mg q12h	500 mg q8h	
IMIPENEM-RELEBACTAM (IV)	**CrCl ≥90 ml/min**	**CrCl 60-89 ml/min**	**CrCl 30-59 ml/min**	**CrCl 15-29 ml/min**	**iHD**	**CRRT**
	1.25 g q6h	1 g q6h	0.75 g q6h	0.5 g q6h	0.5 g q6h	1.25g q8h

ANTIBIOTICS	CrCl[1] > 50 mL/min	CrCl[1] 30-49 mL/min	CrCl[1] 10-29 mL/min	CrCl[1] < 10 mL/min; HD[2]	CRRT (rate 2.5 L/hr)[3]
LEVOFLOXACIN (IV/PO)					
Uncomplicated cystitis (lower urinary tract infection)	500 mg x1, then 250 q24h	500 mg x1, then 250 q24h	500 mg x1, then 250 mg q48h	500 mg x1, then 250 mg q48h	500 mg x1, then 250 q24h
All other infections, including bacteremia, pyelonephritis, pneumonia, and other systemic infections	750 mg q24h	500 mg q24h	750 mg x1, then 250 mg q24h	750 mg x1, then 250 mg q24h	500 mg q24h
LINEZOLID (IV/PO)					
Mycobacterial infections	600mg daily – q12h				
All other infections	600mg q12h				
MEROPENEM-VABORBACTAM (IV)	**eGFR (mL/min/1.73m^2) = 175 * (serum creatinine)$^{-1.154}$ * (age)$^{-0.203}$ * (0.742 if female) * (1.212 if African American)**				
	≥ 50 mL/min/1.73m^2	**49-30 mL/min/1.73m^2**	**29-15 mL/min/1.73m^2**	**< 15 mL/min/1.73m^2**	
All indications	4 g (mero 2 g - vabor 2 g) q8h	2 g (mero 1 g - vabor 1 g) q8h	2 g (mero 1 g - vabor 1 g) q12h	1 g (mero 500 mg - vabor 500 mg) q12h	
MEROPENEM (IV)	Extended infusion over 3 hours is preferred for most patients				
Suspected or proven *Acinetobacter* or *Pseudomonas* infection, febrile neutropenia, CNS infections, severe sepsis and septic shock, burn	2 g q8h	2 g q12h	1 g q12h	1 g q24h	2 g q12h
All other infections	1 g q8h	1 g q12h	500 mg q12h	500 mg q24h	1 g q12h
METRONIDAZOLE (IV/PO)					
CNS infections; *C. difficile* adjunctive therapy	500mg q8h				
Intra-abdominal infections; all other infections	500mg q12h				
MINOCYCLINE (IV/PO)					
Acinetobacter, *Stenotrophomonas* or other infection caused by non-lactose fermenting Gram negative rods	200 mg q12h				
All other infections	200 mg x1, then 100 mg q12h				
MOXIFLOXACIN (IV/PO)	400 mg q24h				
OMADACYCLINE (IV)	200 mg x1, then 100 mg q24h				
OMADACYCLINE (PO)	450 mg q24h x2, then 300 mg q24h				
OXACILLIN/NAFCILLIN (IV)	2 g q4h				
NITROFURANTOIN (capsules – cystitis only)	100 mg q12h		Not recommended		
NITROFURANTOIN (suspension – cystitis only)	50-100 mg q6h		Not recommended		
PENICILLIN G (IV)					
Viridans group streptococci native valve endocarditis (penicillin MIC ≤ 0.12 mg/L)	3 million units q4h	3 million units q4h	3 million units q6h	1.5 million units q6h	3 million units q4h
Viridans group streptococci native valve endocarditis (penicillin MIC > 0.12 mg/L), enterococcal endocarditis, prosthetic valve endocarditis, or neurosyphilis	4 million units q4h	4 million units q4h	4 million units q6h	2 million units q6h	4 million units q4h

ANTIBIOTICS	CrCl[1] > 50 mL/min	CrCl[1] 30-49 mL/min	CrCl[1] 10-29 mL/min	CrCl[1] < 10 mL/min; HD[2]	CRRT (rate 2.5 L/hr)[3]
PIPERACILLIN-TAZOBACTAM (IV)	Extended infusion over 3 hours is preferred for most patients				
Uncomplicated cystitis (lower urinary tract infection)	4.5 g q8h	4.5 g q8h	4.5 g q12h	4.5 g q12h	4.5 g q8h
All other infections	4.5 g q6h	4.5 g q8h	4.5 g q8h	4.5 g q12h	4.5 g q8h
POLYMYXIN B (IV) *	2 mg/kg x1 (max: 200 mg) then 100 mg q12h				
RIFAMPIN (IV/PO)					
Non-cavitary, non-tuberculous Mycobacterial pulmonary disease	600 mg 3x/week				
Mycobacterial infection	450 mg q24h (< 50 kg) or 600 mg q24h (≥ 50 kg)				
Prosthetic valve Staphylococcal endocarditis	300 mg q8h				
SULBACTAM-DURLOBACTAM (IV)	**CrCl[1] > 130mL/min**	**CrCl[1] 45 – 129 mL/min**	**CrCl[1] 30 – 44 mL/min**	**CrCl[1] 15 – 29 mL/min**	**CrCl[1] < 15 mL/min; HD[2]**	**CRRT (rate 2.5 L/hr)[3]**
All indications	2g (sul 1g – dur 1g) q4h	2g (sul 1g – dur 1g) q6h	2g (sul 1g – dur 1g) q8h	2g (sul 1g – dur 1g) q12h	2g (sul 1g – dur 1g) q12h x 3 doses, then 2g (sul 1g – dur 1g) q24h If CrCl declines to < 15 mL/min while already on sulbactam-durlobactam: 2g (sul 1g – dur 1g) q24h	2g (sul 1g – dur 1g) q6h
TIGECYCLINE (IV)					
Acinetobacter, *Stenotrophomonas* or other infection caused by non-lactose fermenting Gram negative rods	100 mg IV q12h				
Atypical mycobacteria	50 mg IV q24h				
Other infections	100 mg x1, then 50 mg q12h				
TOBRAMYCIN (IV)	[See Aminoglycoside Dosing in Adult Patients]				
TRIMETHOPRIM-SULFAMETHOXAZOLE (IV/PO)[4, *]					
Uncomplicated cystitis (lower urinary tract infection)	1 DS tab q12h	1 DS tab q12h	1 DS tab q12-24h	1 DS tab q24h	1 DS tab q12h
Skin & soft tissue	1-2 DS tab q12h	1-2 DS tab q12h	1-2 DS tab q12-24h	1-2 DS tab q24h	1-2 DS tab q12h
Pneumocystis pneumonia, *Nocardia*¸*Toxoplasma*	5 mg/kg IV q8h – q12h (consider limiting to 320 mg/dose)	5 mg/kg IV q8h – q12h (consider limiting to at 320 mg/dose)	5 mg/kg IV q12h – q24h (consider limiting to at 320 mg/dose)	5 mg/kg q24h (consider limiting to at 320 mg/dose)	5 mg/kg IV q12h (consider limiting to at 320 mg/dose)
Listeria meningitis	5 mg/kg q8h	5 mg/kg q8h	5 mg/kg q12h	5 mg/kg q24h	5 mg/kg q12h
All other infections, including bacteremia, pyelonephritis, pneumonia, and other systemic infections	2 DS tab q12h	2 DS tab q12h	2 DS tab q12-24h	2 DS tab q24h	2 DS tab q12h
VANCOMYCIN (IV)	[See Vancomycin Nomogram]				

ANTIFUNGALS	CrCl[1] >50 mL/min	CrCl[1] 30-49 mL/min	CrCl[1] 10-29 mL/min	CrCl[1] <10 mL/min; HD[2]	CRRT (rate 2.5 L/hr)[3]
AMPHOTERICIN B DEOXYCHOLATE (IV) *	0.75-1 mg/kg q24h				
LIPOSOMAL AMPHOTERICIN B (AMBISOME) *					
Systemic infection	3-5 mg/kg q24h				
Mucormycosis	5-10 mg/kg q24h				
FLUCONAZOLE (IV/PO)					
Urinary tract infection or oropharyngeal thrush	200 mg q24h	200 mg x1, then 100 mg q24h	200 mg x1, then 100 mg q24h	200 mg x1 then 100mg q24h or 200mg post HD 3x/week	400 mg q24h
Systemic infections (not listed below), definitive therapy for candidemia	800 mg x1, then 400 mg q24h	800 mg x1, then 200 mg q24h	800 mg x1, then 200 mg q24h	800 mg x1, then 200 mg q24h or 400 mg post HD 3x/week	800 mg q24h
Non-urinary Candida glabrata *infection,* Candida *endocarditis,* Candida *endophthalmitis*	1,600 mg x1, then 800 mg q24h	1,600 mg x1 load, then 400 mg q24h	1,600 mg x1 load, then 400 mg q24h	1,600 mg x1 load, then 400 mg q24h or 800 mg post HD 3x/week	1,600 mg q24h
Cryptococcal meningitis, induction dosing	1,200 mg q24h	1,200 mg X 1, then 600 mg q24h	1,200 mg X 1, then 600 mg q24h	1,200 mg X 1, then 600 mg q24h or 1200 mg post HD 3x/week	Contact ID PharmD
FLUCYTOSINE (PO)*	25 mg/kg q6h	25 mg/kg q8-12h	25 mg/kg q24h	25 mg/kg post HD	25 mg/kg q8-12h
ISAVUCONAZOLE (IV/PO) Dosing expressed in mg of isavuconazonium sulfate (prodrug of isavuconazole)	**Maintenance dosing may vary from below based on therapeutic drug monitoring** 372 mg q8h x6, then 372 mg q24h starting 12h after load				
ITRACONAZOLE (PO CAPSULE AND LIQUID)	200 mg q8h x9, then 200 mg q12h				
ITRACONAZOLE (TOLSURA CAPSULE)	130 mg q8h x9, then 130 mg q12h				
MICAFUNGIN (IV)					
Prophylaxis: primary ALL and pre-engraftment HSCT recipients	50 mg q24h				
Treatment: Invasive candidiasis Prophylaxis: primary AML, relapsed/refractory ALL, and post-engraftment HSCT recipients	100 mg q24h				
Invasive aspergillosis, esophageal candidiasis, infective endocarditis	150 mg q24h				
POSACONAZOLE (PO tablet or IV)	Maintenance dosing may vary from below based on therapeutic drug monitoring				
Prophylaxis or treatment	300 mg q12h x2, then 300 mg q24h				
POSACONAZOLE (PO IR suspension)	Maintenance dosing may vary from below based on therapeutic drug monitoring				
Prophylaxis	200 mg q8h				
Treatment	200 mg q6h				
REZAFUNGIN (IV)	400 mg load on Day 1 followed 200 mg weekly starting on Day 8				
VORICONAZOLE (IV)	Maintenance dosing may vary from below based on therapeutic drug monitoring				
Prophylaxis	200 mg q12h				
Treatment	6 mg/kg q12h x2, then 4 mg/kg q12h				
VORICONAZOLE (PO)	Maintenance dosing may vary from below based on therapeutic drug monitoring				
Prophylaxis	200 mg q12h				
Treatment	6 mg/kg q12h x2, then 4 mg/kg q12h				

ANTIVIRALS	CrCl[1] > 50 mL/min	CrCl[1] 30-49 mL/min	CrCl[1] 10-29 mL/min	CrCl[1] < 10 mL/min; HD[2]	CRRT (rate 2.5 L/hr)[3]
ACYCLOVIR (IV)*					
Herpes Simplex Prophylaxis	200 mg q12h	200 mg q12h	200 mg q24h	100 mg q24h	200 mg q12h
Mild/moderate HSV mucocutaneous infection and genital infection	5 mg/kg q8h	5 mg/kg q12h	5 mg/kg q24h	2.5 mg/kg q24h	5 mg/kg q12h
Severe HSV mucocutaneous infection and genital infection	5 or 10 mg/kg q8h	5 or 10 mg/kg q12h	5 or 10 mg/kg q24h	2.5 or 5 mg/kg q24h	5 or 10 mg/kg q12h
Encephalitis; meningitis; VZV infection	10 mg/kg q8h	10 mg/kg q12h	10 mg/kg q24h	5 mg/kg q24h	10 mg/kg q12h
ACYCLOVIR (PO)					
Herpes Simplex Prophylaxis	400 mg q12h	400 mg q12h	400 mg q12h	400 mg q24h	400 mg q12h
Herpes Zoster	800 mg 5x/day	800 mg q6h	800 mg q8h	800 mg q12h	800 mg q6h
Herpes labialis	400 mg q8h	400 mg q8h	400 mg q12h	400 mg q12h	400 mg q8h
Genital herpes, initial episode	400 mg q8h	400 mg q8h	400 mg q12h	400 mg q12h	400 mg q8h
Genital herpes, recurrence	400 mg q8h	400 mg q8h	400 mg q12h	400 mg q12h	400 mg q8h
BALOXAVIR (PO)					
Treatment for patients 40 kg to <80kg	40 mg x1				
Treatment for patients ≥80 kg	80 mg x1				
CIDOFOVIR (IV)*					
BK cystitis	1 mg/kg x1 then discuss subsequent dosing with ID pharmacy/consultation (no probenecid)				
All other indication (e.g., ganciclovir-resistant CMV, adenovirus)	5 mg/kg x1; then discuss subsequent dosing with ID pharmacy/consult service (administer with probenecid 2 g 3 hours prior to cidofovir dose, then 1 g at 2 hours and 8 hours after the infusion)				

FOSCARNET (IV)*	**mL/min/kg = (140-age)/(72*SCr); multiply by 0.85 for females**							
	≥ 1.4 mL/min/kg	**1.0 - < 1.4 mL/min/kg**	**0.8 - < 1.0 mL/min/kg**	**0.6 - < 0.8 mL/min/kg**	**0.5 - < 0.6 mL/min/kg**	**0.4 - < 0.5 mL/min/kg**	**< 0.4 mL/min/kg**	**HD**
Treatment for CMV or HHV6	90 mg/kg q12h	70 mg/kg q12h	50 mg/kg q12h	80 mg/kg q24h	60 mg/kg q24h	50 mg/kg q24h	No data	60mg/kg post HD
Primary or secondary prophylaxis for CMV or HHV6	90 mg/kg q24h	70 mg/kg q24h	50 mg/kg q24h	80 mg/kg q48h	60 mg/kg q48h	50 mg/kg q48h	No data	60mg/kg post HD
Treatment for HSV or VZV	60 mg/kg q12h	47mg/kg q12h	34mg/kg q12h	54mg/kg q24h	40mg/kg q24h	34mg/kg q24h	No data	40mg/kg post HD
Primary or secondary prophylaxis for HSV or VZV	60 mg/kg q24h	47mg/kg q24h	34mg/kg q24h	54mg/kg q48h	40mg/kg q48h	34mg/kg q48h	No data	40mg/kg post HD

GANCICLOVIR (IV)*					
Treatment	5 mg/kg q12h	2.5 mg/kg q24h	1.25 mg/kg q24h	1.25 mg/kg post HD 3x/week	2.5 mg/kg q12h
Primary or secondary prophylaxis	5 mg/kg q24h	1.25 mg/kg q24h	0.625 mg/kg q24h	0.625 mg/kg post HD 3x/week	2.5 mg/kg q24h
Renal, liver, pancreas transplant prophylaxis	See transplant guidelines				
LETERMOVIR (PO/IV)	480 mg q24h 240 mg q24h (if on cyclosporine)				
MARIBAVIR (PO)	400 mg q12h				

ANTIVIRALS	CrCl[1] > 50 mL/min	CrCl[1] 30-49 mL/min	CrCl[1] 10-29 mL/min	CrCl[1] < 10 mL/min; HD[2]	CRRT (rate 2.5 L/hr)[3]
MOLNUPIRAVIR (PO)	800 mg q12h x5 days				
NIRMATRELVIR-RITONAVIR (PO)	MiChart eGFR ≥ 60 mL/min		MiChart eGFR ≥ 30 - < 60 mL/min	MiChart eGFR < 30 mL/min, HD, CRRT	
1 nirmatrelvir tablet = 150 mg 1 ritonavir tablet = 100 mg	2 nirmatrelvir tablets, 1 ritonavir tablet q12h x5 days		1 nirmatrelvir tablet, 1 ritonavir tablet q12h x5 days	Not recommended	
OSELTAMIVIR (PO)					
Prophylaxis	75 mg q24h	75 mg q24h	75 mg q48h	75 mg 2x/week	75 mg q24h
Treatment	75 mg q12h	75 mg q12h	75 mg q24h	75 mg q48h	75 mg q12h
PERAMIVIR (IV)	600 mg q24h	200 mg q24h	100 mg q24h	100 mg post HD	600 mg q24h
RIBAVIRIN (PO) (Treatment of Respiratory Syncytial Virus in Hematology patients)	600 mg q8h	400 mg q8h	200 mg q8h	200 mg q24h	400 mg q8h
RIBAVIRIN (PO) Lung Transplant recipients *	**≥ 20 mL/min**	**< 20 mL/min**		**HD**	**CRRT**
Treatment of Respiratory Syncytial Virus in Lung Transplant recipients	8 mg/kg q6h x48 hours, then 8 mg/kg q24h	8 mg/kg q6h x48 hours, then 4 mg/kg q24h		8 mg/kg q6h x48 hours, then 200 mg q24h	No data
REMDESIVIR (IV)					
Treatment, severe COVID-19	200 mg IV x1, then 100 mg IV q24h x4 doses (5 days total)				
Treatment, mild/moderate COVID-19	200 mg IV x1, then 100 mg IV q24h x2 doses (3 days total)				
VALACYCLOVIR (PO)					
Herpes labialis	2 g q12h	1 g q12h	500 mg q12h	500 mg x1	500 mg q12h
Herpes zoster	1 g q8h	1 g q12h	1 g q24h	500 mg q24h	1 g q24h
Genital herpes, initial episode	1 g q12h	1 g q12h	1 g q24h	500 mg q24h	1 g q24h
Genital herpes, recurrence	1 g q24h	1 g q24h	500 mg q24h	500 mg q24h	500 mg q24h

VALGANCICLOVIR (PO)	CrCl ≥ 60 mL/min	CrCl 40-59 mL/min	CrCl 25-39 mL/min	CrCl 10-24 mL/min	CrCl < 10 mL/ min or HD	CRRT (rate 2.5 L/hr)
Treatment	900 mg q12h	450 mg q12h	450 mg q24h	450 mg q48h	450 mg post HD 3x/week	450 mg q12h
Primary or secondary prophylaxis	900 mg q24h	450 mg q24h	450 mg q48h	450 mg 2x/week	450 mg 2x/week if not on HD or 3x/week post HD	450 mg q24h
Renal, liver, pancreas transplant prophylaxis	See transplant transplant guidelines					
ZANAMIVIR (INHALED)						
Prophylaxis	10 mg (2 inhalations) q24h					
Treatment	10 mg (2 inhalations) q12h					

Dosing recommendations may fall outside manufacturer guidelines.

1. Cockcroft-Gault; $CrCl\ (\frac{mL}{min}) = \frac{(140-Age)*Weight\ (kg)}{72*SCr(mg/dL)} * (0.85\ for\ females)$ *For weight*, Actual body weight should be used for non-overweight patients.

   For overweight patients (defined as BMI >/= 25), use adjusted body weight.

   $Ideal\ body\ weight\ (male) = 50 + (2.3 * height\ in\ inches > 60\ inches)$; $Ideal\ body\ weight\ (female) = 45 + (2.3 * height\ in\ inches > 60\ inches$;

   $Adjusted\ body\ weight = 0.4(actual\ body\ weight - ideal\ body\ weight) + ideal\ body\ weight$
2. Schedule dose after hemodialysis; if dosed multiple times daily, schedule subsequent doses after hemodialysis.
3. Dosing may be different for more aggressive CRRT flow rates. Consult clinical pharmacist for dosing recommendations. The effluent rate can be used as an incomplete surrogate for drug clearance with CRRT. A general calculation for effluent rate is (these parameters can be found in the CRRT flowsheet): dialysate ml/hr + replacement ml/hr + citrate rate ml/hr + 100 ml/hr (ultrafiltrate estimate, can fluctuate hourly). It is also important to take patient specific factors (i.e. Indication for treatment, patient acuity, drug efficacy and safety parameters) into consideration when near dosing thresholds.
4. Based on trimethoprim component.
5. Manufacturer recommendations for cefiderocol dosing in CRRT:

Effluent Flow Rate[3]	Dose
2 L/hr or less	1.5 g q12h
2.1 to 3 L/hr	2 g q12h
3.1 to 4 L/hr	1.5 g q8h
4.1 L/hr or greater	2 g q8h

6. Cefepime dosing: could consider higher dose of 2 g q8h in difficult to treat infections with elevated MIC's in discussion w/ ID pharmacy

* For dosing in obese patients, please see Weight-Based Dosing Recommendations for Intravenous Antimicrobials in Obese Adult Patients

Last revised: 10/2026	**Antimicrobial Stewardship Subcommittee:** 10/2026	**P&T:** 11/2026

References	
	Citation

Revision History	
Date	**Revision**
01/2020	Added eravacycline and changed CrCl recommendations.
08/2020	Added cefiderocol and imipenem-relebactam
12/2020	Updated ceftazidime and ceftazidime-avibactam dosing
03/2021	Updated vancomycin hyperlink
04/2021	Added SUBA-itraconazole and omadacycline, added cefiderocol CRRT dosing
05/2021	Added tigecycline, updated minocylcine dosing, updated ertapenem HD dosing
06/2021	Adjusted meropenem dosing criteria, adjusted minocycline dosing
07/2021	Adjusted cefepime and meropenem dosing criteria
03/2022	Added clarithromycin, maribavir, molnupiravir, nirmatrelvir-ritonavir, and remdesivir
09/2022	Adjusted ceftriaxone categories
03/2023	Revised aztreonam, cefepime, cefiderocol, cephalexin, imipenem, imipenem-relebactam, meropenem, piperacillin-tazobactam, fluconazole, and posaconazole
08/2023	Revised antifungal sections
12/2023	Added dalbavancin and rezafungin
12/2024	Revised cystitis wording, amoxicillin, amoxicillin-clavulanate, ampicillin, ampicillin-sulbactam, azithromycin, aztreonam, cefepime, cefoxitin, cefpodoxime, ceftaroline, ceftazidime-avibactam, ceftriaxone, cefuroxime, cephalexin, ciprofloxacin, daptomycin, doxycycline, imipenem, linezolid, meropenem, metronidazole, minocycline, piperacillin-tazobactam, sulbactam-dulobactam, trimethoprim-sulfamethoxazole, fluconazole, foscarnet, ganciclovir, and valganciclovir
07/2025	Updated amikacin, gentamicin, tobramycin, vancomycin, and dosing in obesity hyperlinks
11/2025	Added aztreonam-avibactam
05/2026	Revised cephalexin cystitis dosing
10/2026	Revised cefepime dosing